

[ARCHIVED CATALOGUE]

Electrical Engineering (Wireless Networks) (MS)

The Master of Science in Electrical Engineering (Wireless Networks) is a unique interdisciplinary degree program that prepares graduates for the design and improvement of future wireless networks such as the “Internet of Things.” The program combines courses related to radio hardware, transmission techniques, the medium-access control layer, networking, applications and standards.

No more than three courses (maximum 12 units) may be counted at the 400 level — at least 18 adviser-approved units must be taken at the 500 or 600 level. Units to be transferred (maximum 4 with adviser approval) must have been taken prior to taking classes at USC — interruption of residency is not allowed.

For Admission Requirements, refer to [USC Viterbi School of Engineering](#).

The Master of Science in Electrical Engineering (Wireless Networks) requires at least 28 units.

Required Courses (16 Units)

- [CSCI 402 Operating Systems](#) Units: 4
- [EE 503 Probability for Electrical and Computer Engineers](#) Units: 4
- [EE 535 Wireless Communications](#) Units: 4
- [EE 597 Wireless Networks](#) Units: 4

Elective Courses (12-14 Units, at Least One Course from Two Areas)

Transmission Techniques and Signal Processing

- [EE 483 Introduction to Digital Signal Processing](#) Units: 4
- [EE 558 Optical Fiber Communication Systems](#) Units: 3
- [EE 564 Digital Communication and Coding Systems](#) Units: 4
- [EE 583 Statistical Signal Processing](#) Units: 3
- [EE 586L Advanced DSP Design Laboratory](#) Units: 4

Architectures, Protocols, and Applications

- [EE 519 Speech Recognition and Processing for Multimedia](#) Units: 3
- [EE 532 Wireless Internet and Pervasive Computing](#) Units: 3
- [EE 550 Data Networks: Design and Analysis](#) Units: 4
- [EE 555 Broadband Network Architectures](#) Units: 3

Communication Hardware and Design

- [EE 448L Communication Electronics](#) Units: 4
- [EE 541 A Computational Introduction to Deep Learning](#) Units: 2
- [EE 544 Radio Frequency Systems and Hardware](#) Units: 3

Note:

This program assumes prerequisite preparation in the area of computer networks. Students who do not meet this requirement or who do not pass a related placement exam will be required to take [EE 450 Introduction to Computer Networks](#).
